

Package ‘GTAPViz’

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Title Automating GTAP Data Processing and Visualization

Version 0.0.0.9000

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Description GTAPViz is an extension package of HARplus, designed to provide seamless data visualization for GTAP model results from ``.HAR`` and ``.SL4`` files. This package focuses on user-friendly visualization tools, enabling GTAP users to analyze and compare simulation results efficiently without requiring deep coding knowledge. With built-in functions for pivoting, mapping, unit assignments, and graphical representation, GTAPViz enhances economic model interpretation by generating structured, publication-ready plots. It is ideal for GTAP researchers, policymakers, and analysts looking for intuitive data exploration and visual insights from GEMPACK outputs.

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Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

BugReports <https://github.com/bodysbobb/GTAPViz/issues>

URL <https://bodysbobb.github.io/GTAPViz/>

Imports HARplus,

stats,
ggplot2,
dplyr,
tidyr,
openxlsx,
colorspace,
grDevices,
rlang,
scales,
purrr,
readxl,
writexl,
utils,
methods

Remotes github::Bodysbobb/HARplus

Suggests knitr,
rmarkdown,
devtools

VignetteBuilder knitr

Exports add_unit_col,
comparison_plot,
detail_plot,
macro_plot,
gtap_macros_data,
add_mapping_info

ImportFrom HARplus get_data_by_var load_sl4x
stats setNames
ggplot2 ggplot aes aes_string geom_bar geom_text geom_vline geom_col geom_hline facet_wrap
theme_minimal theme element_text element_rect element_blank unit margin labs
scale_fill_manual scale_y_continuous position_dodge expansion
dplyr filter mutate case_when
colorspace hex2RGB hex polarLUV sRGB
grDevices col2rgb rgb
rlang sym
scales squish

Depends R (>= 3.5)

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add_mapping_info	<i>Add Mapping Information to GTAP Data</i>
------------------	---

Description

Adds description and unit information to GTAP data based on specified mapping.

Usage

```
add_mapping_info(  
  data_list,  
  external_map = NULL,  
  mapping = "GTAPv7",  
  description_info = TRUE,  
  unit_info = TRUE  
)
```

Arguments

data_list	Data structure containing GTAP variables
external_map	Optional data frame with mapping information
mapping	Mapping mode: "GTAPv7" (default), "No", "Yes", or "Mix"
description_info	Logical. If TRUE, adds description information
unit_info	Logical. If TRUE, adds unit information

Value

Data structure with added mapping information, preserving original structure

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Load example data
har_data <- HARplus::load_harx(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPviz"))

# Corrected example with closing quotation
har_data <- add_mapping_info(har_data, mapping = "GTAPv7")

## End(Not run)
```

comparison_plot	<i>Generate Comparison Plots</i>
-----------------	----------------------------------

Description

Creates bar plots comparing values across categories with flexible panel configuration.

Usage

```
comparison_plot(
  data,
  plot_var = NULL,
  x_axis_from,
  split_by,
  title_prefix = "",
  title_suffix = "",
  compare_by_experiment = FALSE,
  separate_figure = FALSE,
  invert_panel = FALSE,
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  output_dir = NULL,
```

```

    panel_rows = NULL,
    panel_cols = NULL,
    color_tone = NULL,
    legend_position = "none",
    width = NULL,
    height = NULL
  )

```

Arguments

<code>data</code>	Data frame containing "Variable", "Value", "Experiment", "Unit" columns plus <code>x_axis_from</code> and <code>split_by</code> variables.
<code>plot_var</code>	Optional. Character vector or data frame with "Variable" column to specify which variables to plot.
<code>x_axis_from</code>	Column name to use for x-axis values (or panels if <code>compare_by_experiment = TRUE</code>).
<code>split_by</code>	Column name for separating plots (or panels if <code>invert_panel = TRUE</code>).
<code>title_prefix</code>	Optional text to prepend to plot titles.
<code>title_suffix</code>	Optional text to append to plot titles.
<code>compare_by_experiment</code>	If TRUE, Experiment is plotted on x-axis, <code>x_axis_from</code> values as panels.
<code>separate_figure</code>	If TRUE, creates individual figures for each panel value.
<code>invert_panel</code>	If TRUE, uses <code>split_by</code> for panels and Experiment for separation.
<code>var_name_by_description</code>	If TRUE, uses "Description" column for titles instead of "Variable".
<code>output_dir</code>	Optional path to save plots. If NULL, plots are only returned.
<code>panel_rows</code>	Optional number of panel rows. Auto-calculated if NULL.
<code>panel_cols</code>	Optional number of panel columns. Auto-calculated if NULL.
<code>color_tone</code>	Optional base color for the plot. NULL uses default ggplot2 colors.
<code>legend_position</code>	Position of legend: "none" (default), "bottom", "top", "left", "right".
<code>width</code>	Optional width in inches. Auto-calculated if NULL.
<code>height</code>	Optional height in inches. Auto-calculated if NULL.

Value

A list of ggplot objects or a single ggplot object depending on settings

Author(s)

Pattawee Puangchit

See Also

[detail_plot](#), [macro_plot](#), [, macro_plot](#), [stack_plot](#)

Examples

```
## Not run:
# Generate comparison plot
comparison_plot(
  data = sl4.plot.data$REG,
  x_axis_from = "Region",
  plot_var = sl4plot,
  title_prefix = "Impact on",
  output_dir = output.folder,
  compare_by_experiment = FALSE,
  var_name_by_description = TRUE,
  separate_figure = FALSE,
  color_tone = "grey",
  width = 20,
  height = 12,
  legend_position = "bottom"
)

## End(Not run)
```

convert_units

*Convert Multiple Units in Nested Data Structures***Description**

Convert Multiple Units in Nested Data Structures

Usage

```
convert_units(
  data,
  change_unit_from = NULL,
  change_unit_to = NULL,
  adjustment = NULL,
  value_col = "Value",
  unit_col = "Unit",
  variable_select = NULL,
  variable_col = "Variable",
  scale_auto = NULL
)
```

Arguments

data	A data structure (list, data.frame, or nested combination)
change_unit_from	Character vector of units to change (case-insensitive, space-insensitive, and parentheses-insensitive)
change_unit_to	Character vector of new units (same length as change_unit_from, preserves exactly as entered)
adjustment	Character vector of operations or numeric vector (same length as change_unit_from)

value_col	Column name containing values to adjust (default: "Value")
unit_col	Column name containing unit information (default: "Unit")
variable_select	Optional character vector of variable names to process. If NULL, all variables are processed.
variable_col	Column name containing variable identifiers (default: "Variable")
scale_auto	Optional character vector of predefined conversion rules: "mil2bil", "bil2mil", "pct2frac", "frac2pct"

Details

The function supports both manual and automatic unit conversion:

For manual conversion, provide:

- change_unit_from: Units to convert from (case-insensitive)
- change_unit_to: Target units (preserved exactly as entered)
- adjustment: Conversion operations (e.g., "/1000", "*100")

For automatic conversion, use scale_auto with these options:

- "mil2bil": Convert million USD to billion USD (divides by 1000)
- "bil2mil": Convert billion USD to million USD (multiplies by 1000)
- "pct2frac": Convert percent to fraction (divides by 100)
- "frac2pct": Convert fraction to percent (multiplies by 100)

Value

Data structure with same format as input but with adjusted values and units

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Load example data
har_data <- HARplus::load_harx(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

# Manual conversion: million USD to billion USD
har_data <- convert_units(har_data,
  change_unit_from = c("million USD"),
  change_unit_to = c("billion USD"),
  adjustment = c("/1000"))

# Automatic conversion
har_data <- convert_units(har_data, scale_auto = c("mil2bil", "pct2frac"))

# Convert units only for specific variables
har_data <- convert_units(har_data,
  change_unit_from = c("million USD"),
  change_unit_to = c("billion USD"),
```

```

adjustment = c("/1000"),
variable_select = c("qgdp", "pop"))

## End(Not run)

```

detail_plot

Generate Detailed Plot

Description

Creates a detailed plot reporting all regions or sectors depending on the variable selection. Uses a separate dataframe to specify variables to plot and their titles.

Usage

```

detail_plot(
  data,
  plot_var = NULL,
  y_axis_from,
  split_by,
  title_prefix = "",
  title_suffix = "",
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  panel_rows = NULL,
  panel_cols = NULL,
  positive_color = "#2E8B57",
  negative_color = "#CD5C5C",
  output_dir = NULL,
  top_impact = NULL,
  width = NULL,
  height = NULL,
  separate_figure = FALSE,
  invert_panel = FALSE,
  legend_position = "none",
  width_dodge = 0.5,
  width_bar = 0.4
)

```

Arguments

data	A data frame containing the full dataset.
plot_var	A data frame with "Variable" and "PlotTitle" columns specifying which variables to plot and their titles. If NULL, all variables in the data will be plotted with default titles.
y_axis_from	Character vector. Column names to use for the y-axis.
split_by	Character. Column used to separate different plots.
title_prefix	Optional character string to prepend to the title.
title_suffix	Optional character string to append to the title.

var_name_by_description	Logical. If TRUE, uses the "Description" column from plot_var as the plot title.
panel_rows	Optional numeric. Number of panel rows. If left blank, it will be automatically adjusted.
panel_cols	Optional numeric. Number of panel columns. If left blank, it will be automatically adjusted.
positive_color	Character. Color for positive values.
negative_color	Character. Color for negative values.
output_dir	Optional character. Directory to save the output plot.
top_impact	Optional numeric. Number of top impact factors to include. If left blank, all factors will be included.
width	Optional numeric. Width of the output plot. If left blank, it will be automatically adjusted.
height	Optional numeric. Height of the output plot. If left blank, it will be automatically adjusted.
separate_figure	Logical. If TRUE, creates separate figures for each combination of variable and category.
invert_panel	Logical. If TRUE, swaps the roles of Experiment and split_by in faceting and figure separation.
legend_position	Character. Position of the legend: "none" (default), "bottom", "top", "left", or "right".
width_dodge	Numeric. Dodge width for bar position adjustment. Default is 0.5.
width_bar	Numeric. Width of bars in the plot. Default is 0.4.

Value

A ggplot object or NULL if no plots could be generated.

Author(s)

Pattawee Puangchit

gtap_macros_data

Extract and Aggregate Scalar Macroeconomic Variables

Description

Extracts scalar macroeconomic variables from multiple SL4 datasets and aggregates them into a structured data frame.

Usage

```
gtap_macros_data(
  input_dir = NULL,
  experiment_names = NULL,
  output_dir = NULL,
  output_formats = NULL,
  subtotal_level = FALSE,
  select_var = NULL
)
```

Arguments

<code>input_dir</code>	Character. Directory containing SL4 files.
<code>experiment_names</code>	Character vector. List of experiment names corresponding to SL4 files.
<code>output_dir</code>	Character (optional). Directory to save exported data.
<code>output_formats</code>	Character vector (optional). List of output formats (e.g., "csv", "xlsx").
<code>subtotal_level</code>	Logical. Whether to include subtotal levels in the processed data.
<code>select_var</code>	Character vector (optional). List of specific variable names to filter from the final result. If NULL, all variables are returned.

Value

A sorted dataframe containing processed GTAP macro data.

Author(s)

Pattawee Puangchit

See Also

[add_mapping_info](#), [process_gtap_data](#)

Examples

```
## Not run:
# Method 1: Extract all variables
input_dir <- paste0(input.folder)
experiment <- c("EXP1", "EXP2") # File name (.sl4)
Macros <- gtap_macros_data(input_dir, experiment_names = experiment,
  subtotal_level = FALSE)

# Method 2: Filter specific variables
Macros <- gtap_macros_data(input_dir, experiment_names = experiment,
  select_var = c("qgdp", "pop", "gdpxp"))

## End(Not run)
```

macro_plot

*Generate Macro Plot with Flexible Title Options***Description**

Creates figures of macro variables from the Macros header in the SL4. Uses a separate dataframe to specify variables to plot with options for using either variable names, PlotTitle, or Description columns for plot titles.

Usage

```
macro_plot(
  data,
  plot_var = NULL,
  title_prefix = "",
  title_suffix = "",
  compare_by_experiment = FALSE,
  var_name_by_description = FALSE,
  output_dir = NULL,
  panel_rows = NULL,
  panel_cols = NULL,
  color_tone = NULL,
  separate_figure = FALSE,
  width = NULL,
  height = NULL,
  legend_position = "none"
)
```

Arguments

data	A data frame containing the full dataset with macroeconomic variables.
plot_var	A data frame with "Variable" column and optionally "PlotTitle" or "Description" columns specifying which variables to plot and their titles. If NULL, all variables in the data will be plotted with default titles.
title_prefix	Optional character string to prepend to the title.
title_suffix	Optional character string to append to the title.
compare_by_experiment	Logical. If TRUE, compares experiments within x-axis categories.
var_name_by_description	Logical. If TRUE, uses the "Description" column for plot titles instead of "PlotTitle". Default is FALSE.
output_dir	Optional character. Directory to save the output plot.
panel_rows	Optional numeric. Number of panel rows. If left blank, it will be automatically adjusted.
panel_cols	Optional numeric. Number of panel columns. If left blank, it will be automatically adjusted.
color_tone	Optional character. Base color for the plot. If left blank, a default color will be used.

separate_figure	Logical. If FALSE, combines multiple variables into one plot.
width	Optional numeric. Width of the output plot. If left blank, it will be automatically adjusted.
height	Optional numeric. Height of the output plot. If left blank, it will be automatically adjusted.
legend_position	Position of the legend: "none", "bottom", "top", "left", or "right"

Value

A ggplot object or a list of plots if separate_figure = TRUE.

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
macro_plot(Macros,
  plot_var = macro.map,
  compare_by_experiment = FALSE,
  color_tone = "grey",
  output_dir = output.folder,
  var_name_by_description = FALSE,
  panel_rows = 2,
  width = 45,
  separate_figure = FALSE,
  legend_position = "bottom")

## End(Not run)
```

process_gtap_data

Process GTAP Data Automation with Flexible Output Options

Description

Processes GTAP data from SL4 and HAR files with options for exporting and preparing plot-ready data.

Usage

```
process_gtap_data(
  experiment,
  mapping_info = "GTAPv7",
  project_dir = NULL,
  input_dir = NULL,
  output_dir = NULL,
  output_formats = NULL,
  plot_data = FALSE,
```

```

    region_select = NULL,
    sector_select = NULL,
    subtotal = FALSE,
    sl4_list_name = "sl4.plot.data",
    har_list_name = "har.plot.data",
    sl4_structure_name = "sl4.structure",
    har_structure_name = "har.structure",
    GTAPMacro_name = "GTAPMacro",
    sl4var = NULL,
    harvar = NULL,
    sl4map = NULL,
    harmap = NULL
)

```

Arguments

experiment	Character vector. Case names to process.
mapping_info	Character. Mapping mode: "GTAPv7" (default), "Yes", "No", or "Mix".
project_dir	Character. Path to the project folder with "in" and "out" subfolders.
input_dir	Character. Path to the input folder. Overrides project_dir/in if specified.
output_dir	Character. Path to the output folder. Overrides project_dir/out if specified.
output_formats	Character vector or list. Exports data in these formats (valid: "csv", "stata", "rds", "txt").
plot_data	Logical. If TRUE, prepares data for plotting and assigns to variables.
region_select	Optional vector. Specifies regions to filter the data.
sector_select	Optional vector. Specifies sectors to filter the data.
subtotal	Logical. If TRUE, includes subtotal data. Default is FALSE.
sl4_list_name	Character. Variable name for SL4 plotting data if generating plot data. Default is "sl4.plot.data".
har_list_name	Character. Variable name for HAR plotting data if generating plot data. Default is "har.plot.data".
sl4_structure_name	Character. Variable name for SL4 structure if generating plot data. Default is "sl4.structure".
har_structure_name	Character. Variable name for HAR structure if generating plot data. Default is "har.structure".
GTAPMacro_name	Character. Variable name for GTAP macro data if generating plot data. Default is "GTAPMacro".
sl4var	Data frame, NULL, or FALSE. Variables to extract from SL4 files. Set to NULL to extract all variables, or FALSE to skip SL4 processing.
harvar	Data frame, NULL, or FALSE. Variables to extract from HAR files. Set to NULL to extract all variables, or FALSE to skip HAR processing.
sl4map	Data frame or NULL. Mapping information for SL4 variables (with "Variable", "Description", and "Unit" columns).
harmap	Data frame or NULL. Mapping information for HAR variables (with "Variable", "Description", and "Unit" columns).

Details

This function provides a complete automation workflow for processing GTAP model outputs, with flexible output options and optional filtering for plot data.

The key parameters `sl4var` and `harvar` each accept three different input types:

- **Data frame:** Contains variable mappings with required "Variable" column. When provided, only specified variables will be extracted.
- **NULL:** Extracts all available variables from the respective file type.
- **FALSE:** Completely skips processing of that file type, allowing the function to focus only on the other file type.

The `mapping_info` parameter controls how descriptions and units are assigned:

- **GTAPv7:** Uses standard GTAPv7 definitions (default)
- **Yes:** Uses only the supplied descriptions and units from `sl4map/harmap`
- **No:** Does not add any descriptions or units
- **Mix:** Prioritizes supplied descriptions and units, falling back to GTAPv7 for any missing values

Value

Returns the processed data invisibly, which will not be printed to the console.

Author(s)

Pattawee Puangchit

See Also

[add_mapping_info](#), [gtap_macros_data](#)

Examples

```
## Not run:
# Extract data with region and experiment filters
process_gtap_data(
  sl4var = NULL,
  harvar = NULL,
  sl4map = sl4map,
  harmap = harmap,
  region_select = selected_regions,
  sector_select = NULL,
  subtotal = FALSE,
  experiment = experiment,
  mapping_info = info.mode,
  project_dir = project.folder,
  plot_data = TRUE,
  output_formats = list(
    "csv" = "No",
    "stata" = "No",
    "rds" = "No",
    "txt" = "No"))

## End(Not run)
```

rename_GTAP_bilateral *Rename GTAP Bilateral Trade Columns*

Description

Renames bilateral trade columns in GTAP data output

Usage

```
rename_GTAP_bilateral(data)
```

Arguments

data Data structure containing GTAP bilateral trade data

Value

Data structure with renamed bilateral trade columns

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Load Sample Data
sl4_data <- HARplus::load_sl4x(system.file("extdata/in", "EXP1-WEL.sl4",
                                           package = "GTAPViz"))
sl4_data <- rename_GTAP_bilateral(df)

## End(Not run)
```

rename_value *Rename Values in Data Structures*

Description

Replaces specified values in a column across nested data structures

Usage

```
rename_value(data, column_name = NULL, mapping.file)
```

Arguments

data Data structure containing data to rename
column_name Column name to modify. If NULL, extracted from mapping file
mapping.file Data frame with OldName and NewName columns for renaming

Value

A modified data frame or list of data frames with specified values replaced.

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Example mapping file
wefare.decomp.rename <- data.frame(
  OldName = c("alloc_A1", "ENDWB1", "tech_C1"),
  NewName = c("Allocation", "Endowment", "Technology"),
  ColumnName = "Variable",
  stringsAsFactors = FALSE
)

# Load example data
har_data <- HARplus::load_harx(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

# Apply renaming
modified_data <- rename_value(har_data, mapping.file = wefare.decomp.rename)

## End(Not run)
```

report_table

Generate a Detailed Report Table

Description

Transforms one or more datasets into a wide-format table based on specified grouping columns and pivot column(s), with optional subtotal filtering and export.

Usage

```
report_table(
  data_list,
  pivot_col,
  total_column = FALSE,
  export_table = FALSE,
  separate_file = FALSE,
  output_dir = NULL,
  sheet_names = NULL,
  include_units = FALSE,
  component_exclude = NULL,
  group_by = NULL,
  rename_cols = NULL,
  var_name_by_description = TRUE,
  add_var_info = FALSE,
```

```

    decimal = 2,
    unit_select = NULL,
    separate_sheet_by = NULL,
    subtotal_level = FALSE,
    repeat_label = FALSE,
    workbook_name = "detail_results",
    add_group_line = FALSE
  )

```

Arguments

<code>data_list</code>	A named list of data frames.
<code>pivot_col</code>	A named list indicating which column is pivoted into wide format for each dataset.
<code>total_column</code>	Logical. Adds a "Total" column summing numeric columns if TRUE.
<code>export_table</code>	Logical. Exports the tables to Excel if TRUE.
<code>separate_file</code>	Logical. If TRUE, each data frame is saved in a separate Excel file.
<code>output_dir</code>	Character. Directory where Excel files are written if <code>export_table=TRUE</code> .
<code>sheet_names</code>	Optional named list for custom sheet names.
<code>include_units</code>	Logical. If TRUE, includes the "Unit" column in grouping.
<code>component_exclude</code>	Optional character vector of pivoted values to exclude.
<code>group_by</code>	A list or character vector of grouping columns, or NULL.
<code>rename_cols</code>	A named list for renaming columns.
<code>var_name_by_description</code>	Logical. If TRUE, replaces Variable names with Description or vice versa.
<code>add_var_info</code>	Logical. If TRUE, appends the other part in parentheses.
<code>decimal</code>	Numeric. Decimal places for rounding.
<code>unit_select</code>	Optional. Filters rows to this unit.
<code>separate_sheet_by</code>	Optional column name to split sheets.
<code>subtotal_level</code>	Logical. If TRUE, includes all Subtotal values; otherwise keeps only Subtotal="TOTAL".
<code>repeat_label</code>	Logical. If TRUE, repeats the first group column in exports.
<code>workbook_name</code>	Character. Name of the Excel workbook (no extension).
<code>add_group_line</code>	Logical. If TRUE, draws a thin line after each group.

Value

A named list of data frames, each transformed into wide format, optionally exported to Excel.

Author(s)

Pattawee Puangchit

stack_plot

*Generate Stacked Bar Plot with Auto-Dimensions***Description**

Creates stacked bar plots with automatic total calculations and flexible grouping. Uses a separate dataframe to specify variables to plot and their titles. Auto-adjusts dimensions and panel layout if not specified.

Usage

```
stack_plot(
  data,
  plot_var = NULL,
  x_axis_from,
  stack_value_from,
  title_prefix = "",
  title_suffix = "",
  y_axis_title = NULL,
  compare_by_experiment = FALSE,
  separate_figure = FALSE,
  unstack_plot = FALSE,
  output_dir = NULL,
  panel_rows = NULL,
  panel_cols = NULL,
  color_tone = NULL,
  show_total = TRUE,
  var_name_by_description = TRUE,
  add_var_info = FALSE,
  legend_position = "bottom",
  y_limit = NULL,
  width = NULL,
  height = NULL
)
```

Arguments

data	A data frame containing the full dataset.
plot_var	A data frame with "Variable" and "PlotTitle" columns specifying which variables to plot and their titles. If NULL, all variables in the data will be plotted with default titles.
x_axis_from	Character. Column name to use for x-axis grouping
stack_value_from	Character. Column name containing the stack categories
title_prefix	Optional character string to prepend to the title
title_suffix	Optional character string to append to the title
y_axis_title	Character. Y-axis label. If NULL, uses Unit if available.
compare_by_experiment	Logical. If TRUE, compares x-axis values within experiments

separate_figure	Logical. If TRUE, creates separate figures instead of panels
unstack_plot	Logical. If TRUE, displays components side by side instead of stacked and creates separate figures for each x_axis_from value
output_dir	Optional character. Directory to save the output plot
panel_rows	Optional numeric. Number of panel rows
panel_cols	Optional numeric. Number of panel columns
color_tone	Optional character. Base color for plot color generation
show_total	Logical. If TRUE, displays total values above stacked bars. Default is TRUE.
var_name_by_description	Logical. If TRUE, uses the "Description" column as the title instead of "Variable". Default is TRUE.
legend_position	Character. Position of the legend: "bottom" (default), "top", "left", "right", or "none"
y_limit	Optional numeric vector. Custom y-axis limits (e.g., c(-5, 5))
width	Optional numeric. Width of the output plot
height	Optional numeric. Height of the output plot

Value

A list of ggplot objects or a single ggplot object depending on settings

Author(s)

Pattawee Puangchit

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